

Solving Equations with Square Roots and Fractions

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Some Important Rules

- ▶ $(x + y)^2 = x^2 + 2xy + y^2$
- ▶ $(x - y)^2 = x^2 - 2xy + y^2$
- ▶ $(x + y)(x - y) = x^2 - y^2$
- ▶ $\sqrt{xy} = \sqrt{x}\sqrt{y}, \quad x, y \geq 0$
- ▶ $\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}, \quad x \geq 0, y > 0$
- ▶ $\frac{x + y}{z} = \frac{x}{z} + \frac{y}{z}, \quad z \neq 0$
- ▶ **Quadratic formula:** Solution of the quadratic equation

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Beware the Shortcut

Certain “simplifications” are simply untrue. Here is one of the most egregious examples.

$$\sqrt{x^2 + y^2} \neq x + y$$

Why? What are you really saying if you claim the two are equal?

Squaring both sides:

$$x^2 + y^2 \quad \text{vs.} \quad (x + y)^2$$

$$x^2 + y^2 \quad \text{vs.} \quad x^2 + 2xy + y^2$$

This is only true if x or y is 0.

For example, $\sqrt{3^2 + 2^2} \neq 3 + 2$

An Equation with a Square Root

Strategy: Isolate the square root on one side. Then square both sides. Check final answers by substituting back.

Example. Solve for x : $\sqrt{4x + 7} = x + 2$.

Solution. Square both sides, being careful to square the sum $x + 2$ correctly:

$$4x + 7 = (x + 2)^2 = x^2 + 4x + 4$$

$$\implies x^2 - 3 = 0$$

$$\implies x = \pm\sqrt{3}.$$

$$x = \sqrt{3} \Rightarrow 4x + 7 = 4\sqrt{3} + 7, \quad (x + 2)^2 = x^2 + 4x + 4 = 7 + 4\sqrt{3} \quad \checkmark$$

$$x = -\sqrt{3} \Rightarrow 4x + 7 = -4\sqrt{3} + 7, \quad (x + 2)^2 = x^2 + 4x + 4 = 7 - 4\sqrt{3} \quad \checkmark$$

Two Square Roots

Strategy: Make sure each side has only one square root. Then square both sides until all square roots are eliminated.

Example. Solve for x : $\sqrt{x} + \sqrt{x+5} = 3$.

Solution.

$$\begin{aligned}\sqrt{x} + \sqrt{x+5} = 3 &\implies \sqrt{x} = 3 - \sqrt{x+5} \\ &\implies x = 9 - 6\sqrt{x+5} + x + 5\end{aligned}$$

$$\implies 0 = -6\sqrt{x+5} + 14$$

$$\implies x + 5 = \left(\frac{7}{3}\right)^2$$

$$\implies x = \frac{4}{9}$$

$$x = \frac{4}{9} \implies \sqrt{x} + \sqrt{x+5} = \frac{2}{3} + \sqrt{\frac{49}{9}} = \frac{2}{3} + \frac{7}{3} = 3 \quad \checkmark$$

Three Square Roots

Strategy: Combine or isolate one square root. Square both sides to reduce the number of square roots. Repeat until all square roots are eliminated. Always check for invalid solutions.

Example. Solve for x : $\sqrt{x} + \sqrt{2x+1} = \sqrt{7x-3}$.

Solution. Squaring both sides,

$$\begin{aligned}\sqrt{x} + \sqrt{2x+1} = \sqrt{7x-3} &\implies x + 2\sqrt{x}\sqrt{2x+1} + 2x + 1 = 7x - 3 \\ &\implies 4x - 4 = 2\sqrt{2x^2 + x} \\ &\implies (2x - 2)^2 = 2x^2 + x \\ &\implies 2x^2 - 9x + 4 = 0 \\ &\implies (2x - 1)(x - 4) = 0 \\ &\implies x = \frac{1}{2}, 4.\end{aligned}$$

By inspection, we see that $x = \frac{1}{2}$ is not a solution to the original equation, hence the solution is $x = 4$.

Multiplying by the Conjugate

Example. Simplify the expression: $\frac{1}{\sqrt{7} - \sqrt{3}}$.

Solution. The conjugate of $a + b$ is $a - b$. When we multiply a conjugate pair, we get $(a + b)(a - b) = a^2 - b^2$. This can be useful in simplifying expressions where one or more terms has a square root.

$$\begin{aligned}\frac{1}{\sqrt{7} - \sqrt{3}} &= \frac{1}{\sqrt{7} - \sqrt{3}} \cdot \frac{\sqrt{7} + \sqrt{3}}{\sqrt{7} + \sqrt{3}} \\ &= \frac{\sqrt{7} + \sqrt{3}}{7 - 3} \\ &= \boxed{\frac{1}{4}(\sqrt{7} + \sqrt{3})}\end{aligned}$$

Common Denominators

Example. Solve for x : $\frac{4}{x-1} - \frac{5}{x+1} = 3$.

Solution. Multiply both sides by the common denominator $(x-1)(x+1)$:

$$(x-1)(x+1) \left(\frac{4}{x-1} - \frac{5}{x+1} \right) = 3(x-1)(x+1)$$

$$4(x+1) - 5(x-1) = 3(x^2 - 1)$$

$$4x+4-5x+5 = 3x^2-3 \Rightarrow -x+9 = 3x^2-3 \Rightarrow 0 = 3x^2+x-12$$

$$x = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 3 \cdot (-12)}}{2 \cdot 3} = \frac{-1 \pm \sqrt{145}}{6}$$

Check: $x \neq \pm 1$, so both solutions are valid.

Another Pitfall

Although $\frac{x+y}{z} = \frac{x}{z} + \frac{y}{z}$, $z \neq 0$ (i.e. each term in the numerator can be put into a different fraction), a sum in the denominator cannot be split:

$$\frac{x}{y+z} \neq \frac{x}{y} + \frac{x}{z}.$$

You can plug in simple values to convince yourself of this:

$x = y = z = 1$ gives us $\frac{1}{2} \stackrel{?}{=} 1 + 1$, which is clearly false.

Least Common Denominators examples

Example. Solve for x :

$$\frac{1}{x} + \frac{2}{x+1} = \frac{3}{x+2}$$

Solution: Multiply both sides by the least common denominator $x(x+1)(x+2)$:

$$x(x+1)(x+2) \left(\frac{1}{x} + \frac{2}{x+1} \right) = x(x+1)(x+2) \cdot \frac{3}{x+2}$$

$$(x+1)(x+2) + 2x(x+2) = 3x(x+1)$$

$$(x^2 + 3x + 2) + (2x^2 + 4x) = 3x^2 + 3x$$

$$3x^2 + 7x + 2 = 3x^2 + 3x \Rightarrow 4x + 2 = 0 \Rightarrow x = -\frac{1}{2}$$

Check: $x \neq 0, -1, -2 \rightarrow$ valid solution.

Simplifying Fractions

Example. Write as a single fraction: $\frac{\frac{1}{x+1} - \frac{2}{x+2}}{\frac{3}{x+3}}$.

Solution. Finding a common denominator, we have

$$\begin{aligned} \frac{\frac{(x+2) - 2(x+1)}{x^2 + 3x + 2}}{\frac{3}{x+3}} &= \left(\frac{-x}{x^2 + 3x + 2} \right) \left(\frac{x+3}{3} \right) \\ &= \frac{-x^2 - 3x}{3(x^2 + 3x + 2)} \end{aligned}$$